MATH1025B
Essential Mathematical Methods

# Chapter 7: Integers and Rational Numbers

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# Divisibility

## Definition 7.1

Let $a, b \in \mathbb{Z}$. We say that $a$ **divides** $b$ (or $b$ is **divisible** by $a$), denoted by $a \mid b$, if there exists $d \in \mathbb{Z}$ such that $b = ad$.
In this case, $a$ is a divisor (or factor) of $b$

## Example 7.1

(i) $3 \mid 15$ but $4 \nmid 15$

(ii) $\pm 1, \pm 3, \pm 5, \pm 15 \mid 15$

(iii) $\pm 1, \pm a \mid a$ for any $a \in \mathbb{Z}$

(iv) $a \mid 0$ for any $a \in \mathbb{Z}$ (In particular $0 \mid 0$)

**Example 7.2**

No integer except 0 is divisible by 0.

The statement is formally expressed as:

"Let $x$ be an integer. Suppose $x$ is divisible by 0. Then $x = 0$."

**Justification:**
Let $x$ be an integer. Suppose $x$ is divisible by 0.
By definition, there exists an integer $k$ such that $x = k \cdot 0$.
Then $x = 0$.

**Proposition 7.1**

*Let $a, b, c \in \mathbb{Z}$.*

(1) *If $b \neq 0$ and $a \mid b$, then $|a| \leq |b|$.*

(2) *If $a \mid b$, then $(\pm a) \mid (\pm b)$.*

(3) *If $a \mid b$ and $b \mid c$, then $a \mid c$.*

(4) *If $a \mid b$ and $a \mid c$, then $a \mid (bx + cy)$ for any $x, y \in \mathbb{Z}$.*

**Proof of (1).**
Since $a \mid b$, there exist $m \in \mathbb{Z}$ such that $b = am$. Then $|b| = |a||m|$.
Since $b \neq 0$, we have $|m| \neq 0$ and so $|m| \geq 1$.
Hence $|b| = |a||m| \geq |a| \cdot 1 = |a|$.

**Proof of (4).**
Since $a \mid b$ and $a \mid c$, there exist $m, n \in \mathbb{Z}$ such that $b = am$ and $c = an$.
Then, for any $x, y \in \mathbb{Z}$,

$$bx + cy = a(mx + ny), \qquad \text{where } mx + ny \in \mathbb{Z}.$$

So $a \mid (bx + cy)$.

**Definition 7.2**

Let $n \in \mathbb{Z}^+$ such that $n > 1$. $n$ is a **prime** if 1 and $n$ are the only positive divisors of $n$. Otherwise $n$ is called a **composite**.

**Remark:** 1 is neither a prime nor a composite!

**Reformulation of the definition of composites**
Let $n \in \mathbb{Z}^+$ such that $n > 1$. Then $n$ is a composite if and only if the statement $(*)$ holds:

$$\text{there exist integers } n_1, n_2 \text{ with } 1 < n_1, n_2 < n \text{ such that } n = n_1 n_2. \qquad (*)$$

**Example 7.3**

(a) $2, 3, 5, 7, 11, \ldots$ are primes.

(b) $4, 6, 8, 9, 10, \ldots$ are composites.

**Question:** How many primes are there?
We will show that there are infinitely many primes.

### Definition 7.3

Let $S$ be a subset of $\mathbb{R}$. Let $m \in S$. We say that $m$ is a **least element of** $S$ if

$$\text{for any } s \in S,\ m \leq s.$$

**Remark:** There can be at most one least element because if $m_1, m_2$ are both least element of $S$, then $m_1 \leq m_2$ and $m_2 \leq m_1$, and so $m_1 = m_2$.

### Theorem 7.2 (Well-ordering Principle)

*Let $S$ be a subset of $\mathbb{N} = \{0, 1, 2, \dots\}$. Suppose $S \neq \emptyset$. Then $S$ has a least element, i.e. there exists $m \in S$ such that $m \leq s$ for all $s \in S$.*

In contrast, a nonempty subset of $\mathbb{R}$ may not have a least element.
For example, $(0, 1]$ does not have a least element even though $x > 0$ for any $x \in (0, 1]$.
Note that 0 is a lower bound but not an element of $(0, 1]$.

**Theorem 7.2 (Well-ordering Principle)**

*Let $S$ be a subset of $\mathbb{N} = \{0, 1, 2, \dots\}$. Suppose $S \neq \emptyset$. Then $S$ has a least element, i.e. there exists $m \in S$ such that $m \leq s$ for all $s \in S$.*

We will show that the Well-ordering Principle is a consequence of the Principle of Mathematical Induction.

**Proof.**
Let $S$ be a subset of $\mathbb{N}$ with no least element.
For $n \in \mathbb{N}$, denote by $P(n)$ the statement: $n \notin S$.
Clearly, $P(0)$ is true since 0 would be the least element of $S$ otherwise.
Suppose $P(k)$ is true for all integers $k = 0, 1, \dots, n$.
Now if $n + 1 \in S$, then $n + 1$ would be the least element of $S$ since

$$0, 1, \dots, n \notin S.$$

So $n + 1 \notin S$ and $P(n + 1)$ is true.
By strong induction, $P(n)$ is true for all $n \in \mathbb{N}$, i.e. $S = \emptyset$. $\square$

**Lemma 7.3**

*Every positive integer greater than 1 has a prime factor.*

**Proof.**
Let $S$ be the set of all integers greater than 1 that has no prime factors.
Suppose on the contrary that $S \neq \emptyset$.
By the Well-ordering Principle, $S$ has a least element $m$.
Firstly, $m$ is not a prime. Otherwise $m$ is a prime factor of itself.
Then $m$ must be composite, and so

$$m = n_1 n_2 \qquad \text{for some } n_1, n_2 \in \mathbb{Z} \text{ with } 1 < n_1, n_2 < m.$$

Since $1 < n_1 < m$, we have $n_1 \notin S$ ($\because m$ is the least element).
By definition of $S$, $p \mid n_1$ for some prime $p$.
Now, $p \mid n_1$ and $n_1 \mid m \implies p \mid m$, which is a contradiction. $\square$

**Proposition 7.4**

*There are infinitely many primes.*

**Proof.**
Assume on the contrary that $p_1 < p_2 < \cdots < p_N$ are the only primes.
Consider $n = p_1 p_2 \cdots p_N + 1 > 1$.
Then $n$ must have a prime divisor, say $p_i \mid n$.
Furthermore, $p_i \mid p_1 p_2 \cdots p_N$.
Therefore, $p_i \mid (n - p_1 p_2 \cdots p_N) = 1$, which is a contradiction. $\square$

## Definition 7.4

(1) Let $a, b \in \mathbb{Z}$. $a$ and $b$ are said to be **relatively prime** if 1 is the only positive divisor of both $a$ and $b$.

(2) Let $a, b \in \mathbb{Z}$. The **greatest common divisor** of $a$ and $b$ is defined as

$$\gcd(a,b) = \begin{cases} \max\{d \in \mathbb{Z} : d \mid a \text{ and } d \mid b\} & \text{if } a \neq 0 \text{ or } b \neq 0 \\ 0 & \text{if } a = b = 0. \end{cases}$$

(Therefore, if $a \neq 0$, $\gcd(a, 0) = |a|$.)

**Remark:**

- $\{d \in \mathbb{Z} : d \mid a \text{ and } d \mid b\} \neq \emptyset$ since 1 divides any integer.
- If $a \neq 0$ or $b \neq 0$, and $d \mid a$ and $d \mid b$, then $|d| \leq |a|$ or $|d| \leq |b|$. So $\{d \in \mathbb{Z} : d \mid a \text{ and } d \mid b\}$ consists of finitely many elements and its maximum does exist.
- $\gcd(a, b) \in \mathbb{Z}^+$ unless $a = b = 0$.

**Proposition 7.5**

*Let $a$ and $b$ be nonzero integers.*

(1) *$a$ and $b$ are relatively prime if and only if* $\gcd(a, b) = 1$.

(2) $\gcd(a, b) = \gcd(|a|, |b|)$.

(3) *If $d = \gcd(a, b)$, then* $\gcd(\frac{a}{d}, \frac{b}{d}) = 1$.

**Proof of (3).**
Let $d' = \gcd(\frac{a}{d}, \frac{b}{d}) > 0$. Then

$$\begin{aligned} & d' \mid \frac{a}{d} \quad \text{and} \quad d' \mid \frac{b}{d} \\ \implies & dd' \mid a \quad \text{and} \quad dd' \mid b \\ \implies & 0 < dd' \leq d \\ \implies & d' = 1. \end{aligned}$$

**Proposition 7.6**

*Let $a$ and $b$ be nonzero integers.*

(1) *$a$ and $b$ are relatively prime if and only if* $\gcd(a, b) = 1$.

(2) $\gcd(a, b) = \gcd(|a|, |b|)$.

(3) *If $d = \gcd(a, b)$, then* $\gcd(\frac{a}{d}, \frac{b}{d}) = 1$.

E.g. To simplify $\dfrac{126}{140}$, note that $\gcd(126, 140) = 14$, then

$$\frac{126}{140} = \frac{126/14}{140/14} = \frac{9}{10},$$

where 9 and 10 are now relatively prime.

But how can we find the greatest common divisor in general?

# Division Algorithm

**Theorem 7.7 (Division Algorithm)**

*Let $a, b \in \mathbb{Z}$ with $b \neq 0$. Then, there exist unique integers $q$ and $r$ such that $a = bq + r$ with $0 \leq r < |b|$.*

The numbers $q$ and $r$ are called the **quotient** and **remainder** in the division of $a$ by $b$.

**Idea:**
Suppose $a \geq b > 0$. We subtract $b$ from $a$ repeatedly until we hit the negative:

$$\begin{aligned} a - b &\geq 0 \\ a - 2b &\geq 0 \\ &\vdots \\ a - qb &\geq 0 \\ a - (q+1)b &< 0 \end{aligned}$$

Let $r = a - bq$. Then $r \geq 0$ and $r - b < 0$, i.e. $0 \leq r < b$.

### Well-ordering Principle

*Let $S$ be a subset of $\mathbb{N} = \{0, 1, 2, \dots\}$. Suppose $S \neq \emptyset$. Then $S$ has a least element, i.e. there exists $m \in S$ such that $m \leq s$ for all $s \in S$.*

### Theorem 7.8 (Division Algorithm)

*Let $a, b \in \mathbb{Z}$ with $b \neq 0$. Then, there exist unique integers $q$ and $r$ such that $a = bq + r$ with $0 \leq r < |b|$.*

**Proof of existence part (for the case $a \geq 0$ and $b > 0$).**
Let $S = \{a - bn : n \in \mathbb{N} \text{ and } a - bn \geq 0\}$.
Since $a = a - b \cdot 0 \in S$, $S \neq \emptyset$.
By the Well-ordering Principle, $S$ has a least element $r$.
Since $r \in S$, we have $r \in \mathbb{N}$ (so $r \geq 0$) and there exists $q \in \mathbb{N}$ such that $r = a - bq$. We now verify that $r < b$.
If it were true that $r \geq b$, then

$$0 \leq r - b = a - (q + 1)b < a - qb = r.$$

So $a - b(q + 1) \in S$, which contradicts the fact that $a - bq$ is the least element of $S$.

**Theorem 7.9 (Division Algorithm)**

*Let $a, b \in \mathbb{Z}$ with $b \neq 0$. Then, there exist unique integers $q$ and $r$ such that $a = bq + r$ with $0 \leq r < |b|$.*

**Proof of uniqueness part.**
Suppose that $a = bq + r = bq' + r'$ where $q, q', r, r' \in \mathbb{Z}$ with $0 \leq r, r' < |b|$.
Then $b(q - q') = r' - r$, and thus $b \mid (r' - r)$.
In particular, $|b| \leq |r' - r|$ unless $r' - r = 0$.
Since $0 \leq r, r' < |b|$, we have

$$-|b| < r' - r < |b|$$

Hence we must have $r' - r = 0$, i.e. $r = r'$.
As a result, $b(q - q') = 0$.
Since $b \neq 0$, we have $q - q' = 0$, i.e. $q = q'$. $\square$

**Exercise 7.1**

(a) Prove the Division Algorithm for $a, b \in \mathbb{Z}$ with $b < 0$.

(b) Prove the Division Algorithm for $a, b \in \mathbb{Z}$ with $a < 0$ and $b > 0$.

**Hint:** If $a < 0$ and $b > 0$, then $a + (-a)b = (-a)(b-1) \in \mathbb{N}$. Apply the Division Algorithm.

**Proposition 7.10**

*Let $a, b, q, r \in \mathbb{Z}$ such that $a = bq + r$. Then $\gcd(a, b) = \gcd(b, r)$.*

**Proof.**
Let $d = \gcd(a, b)$ and $d' = \gcd(b, r)$. Then

$$\begin{aligned} d' \mid b \text{ and } d' \mid r &\implies d' \mid a = bq + r \text{ and } d' \mid b \\ &\implies d' \text{ is a common factor of } a, b \\ &\implies d' \leq d, \end{aligned}$$

and

$$\begin{aligned} d \mid a \text{ and } d \mid b &\implies d \mid r = a - bq \text{ and } d \mid b \\ &\implies d \text{ is a common factor of } b, r \\ &\implies d \leq d'. \end{aligned}$$

So $\gcd(a, b) = d = d' = \gcd(b, r)$. $\square$

**Example 7.4**

Find $\gcd(7224, 1290)$.

$$
\begin{aligned}
7224 &= 1290 \times 5 + 774 & \gcd(7224, 1290) &= \gcd(1290, 774) \\
1290 &= 774 \times 1 + 516 & &= \gcd(774, 516) \\
774 &= 516 \times 1 + 258 & &= \gcd(516, 258) \\
516 &= 258 \times 2 + 0 & &= \gcd(258, 0) = 258
\end{aligned}
$$

It is called the **Euclidean algorithm.**

$$
\begin{aligned}
258 &= 774 + 516 \times (-1) \\
&= 774 + [1290 + 774 \times (-1)] \times (-1) = 1290 \times (-1) + 774 \times 2 \\
&= 1290 \times (-1) + [7224 + 1290 \times (-5)] \times 2 = 7224 \times 2 + 1290 \times (-11)
\end{aligned}
$$

It is called the **extended Euclidean algorithm.**

**Euclidean Algorithm**

Let $a, b \in \mathbb{Z}^+$. Suppose $a > b$. Set $r_0 = a$, $r_1 = b$.
Applying the Division Algorithm iteratively, we obtain

$$\begin{aligned}
r_0 &= r_1 q_1 + r_2 && (0 \le r_2 < r_1) \\
r_1 &= r_2 q_2 + r_3 && (0 \le r_3 < r_2) \\
r_2 &= r_3 q_3 + r_4 && (0 \le r_4 < r_3) \\
&\vdots \\
r_{n-2} &= r_{n-1} q_{n-1} + r_n && (0 \le r_n < r_{n-1}) \\
r_{n-1} &= r_n q_n + 0 && (r_{n+1} = 0)
\end{aligned}$$

Here $r_n$ is the last nonzero remainder. Such $r_n$ exists because

$$r_0 > r_1 > r_2 > r_3 > \cdots > r_{n-1} > r_n > 0.$$

Then

$$\gcd(a, b) = \gcd(r_0, r_1) = \gcd(r_1, r_2) = \cdots = \gcd(r_{n-1}, r_n) = \gcd(r_n, 0) = r_n.$$

# Euclidean Algorithm

How to find $x, y \in \mathbb{Z}$ such that $\gcd(a, b) = ax + by$?

**Extended Euclidean Algorithm**

$$\text{Write} \qquad \begin{aligned} r_0 &= 1 \cdot a + 0 \cdot b = s_0 a + t_0 b \\ r_1 &= 0 \cdot a + 1 \cdot b = s_1 a + t_1 b. \end{aligned}$$

If $r_k = s_k a + t_k b$ and $r_{k+1} = s_{k+1} a + t_{k+1} b$, then

$$\begin{aligned} r_{k+2} &= r_k - q_{k+1} r_{k+1} \\ &= (s_k a + t_k b) - q_{k+1}(s_{k+1} a + t_{k+1} b) \\ &= (s_k - q_{k+1} s_{k+1}) a + (t_k - q_{k+1} t_{k+1}) b. \end{aligned}$$

So we define the integer sequences $\{s_k\}$ and $\{t_k\}$ by

$$\begin{aligned} s_0 &= 1, \quad t_0 = 0 \\ s_1 &= 0, \quad t_1 = 1 \\ s_{k+2} &= s_k - q_{k+1} s_{k+1}, \quad t_{k+2} = t_k - q_{k+1} t_{k+1} \qquad \text{for } k \geq 0. \end{aligned}$$

Then

$$\gcd(a, b) = r_n = s_n a + t_n b.$$

**Example 7.5**

Find $\gcd(12345, 11111)$.

| $k$ | $r_k = r_{k-2} - q_{k-1}r_{k-1}$ | $q_k$ | $s_k$ | $t_k$ |
|---|---|---|---|---|
| 0 | 12345 | | 1 | 0 |
| 1 | 11111 | 1 | 0 | 1 |
| 2 | $12345 - 11111 \times 1 = 1234$ | 9 | 1 | $-1$ |
| 3 | $11111 - 1234 \times 9 = 5$ | 246 | $-9$ | 10 |
| 4 | $1234 - 5 \times 246 = 4$ | 1 | 2215 | $-2461$ |
| 5 | $5 - 4 \times 1 = 1$ | 4 | $-2224$ | 2471 |
| 6 | 0 | | | |

So $\gcd(12345, 11111) = 1 = 12345 \times (-2224) + 11111 \times (2471)$.

**Proposition 7.11 (Bézout's Identity)**

*Let $a, b \in \mathbb{Z}$. Then there exist $x, y \in \mathbb{Z}$ such that $\gcd(a, b) = ax + by$, i.e. $\gcd(a, b)$ can be expressed as a linear combination of $a$ and $b$.*

(If $a = 0$ or $b = 0$, trivial;
if both $a$ and $b$ are nonzero, apply the extended Euclidean algorithm.)

**Direct consequence:**
Let $a, b$ be nonzero integers. Then

$$\begin{aligned} a \text{ and } b \text{ are relatively prime} &\iff \gcd(a, b) = 1 \\ &\implies \text{there exist } x, y \in \mathbb{Z} \text{ such that } ax + by = 1 \\ &\overset{?}{\Longleftarrow} \end{aligned}$$

In fact, it is true!
Let $d = \gcd(a, b)$. Then $d \mid a$ and $d \mid b \implies d \mid 1 = ax + by$, forcing $d = 1$.

**Proposition 7.12**

*Let $a, b$ be nonzero integers. Then $a, b$ are relatively prime if and only if there exist $x, y \in \mathbb{Z}$ such that $ax + by = 1$.*

**Exercise 7.2**

Let $a \in \mathbb{Z}$ and let $p$ be a prime. Show that if $p \nmid a$, then $\gcd(a, p) = 1$. (Equivalently, if $\gcd(a, p) > 1$, then $p \mid a$.)

## Proposition 7.13

*Let $a, b, n \in \mathbb{Z}$. Then $n \mid a$ and $n \mid b$ if and only if $n \mid \gcd(a, b)$.*

**Proof.**
( $\Longrightarrow$ ) It follows directly from Bézout's Identity:

$$\gcd(a, b) = ax + by \text{ for some } x, y \in \mathbb{Z}.$$

( $\Longleftarrow$ ) Clear because $\gcd(a, b) \mid a$ and $\gcd(a, b) \mid b$. $\square$

## Alternative definition of greatest common divisor

*Let $a, b \in \mathbb{Z}$ and let $d \in \mathbb{N}$. The statements (1), (2) are equivalent:*

(1) $d = \gcd(a, b)$

(2) *$d$ is a common divisor of $a$, $b$, and $d$ is divisible by every common divisor of $a, b$.*

**Proposition 7.14 (Euclid's Lemma)**

*Let $a, b \in \mathbb{Z}$ and let $p$ be a prime. If $p \mid ab$, then $p \mid a$ or $p \mid b$.*

**Proof.**
If $p \mid a$, done.
If $p \nmid a$, then $\gcd(a, p) = 1$ and $am + pn = 1$ for some $m, n \in \mathbb{Z}$.
So $abm + bpn = b$.
Since $p \mid ab$, $ab = pq$ for some $q \in \mathbb{Z}$.
Thus $b = pqm + bpn = p(qm + bn)$, where $qm + bn \in \mathbb{Z}$.
Therefore $p \mid b$. $\square$

**Corollary 7.15 (Corollary to Euclid's lemma)**

*Let $p$ be a prime number. Let $n \in \mathbb{Z}^+ \backslash \{1\}$ and let $a_1, a_2, \ldots, a_n \in \mathbb{Z}$.*
*If $p$ divides $a_1 a_2 \cdots a_n$, then $p$ divides at least one of $a_1, a_2, \ldots, a_n$.*

**Theorem 7.16 (Fundamental Theorem of Arithmetic)**

*Every positive integer $> 1$ can be expressed as a product of prime(s) in a unique way. (Primes are "building blocks" of positive integers $> 1$.)*

**Proof of existence part.**
Let
$S = \{n \in \mathbb{Z}^+ \backslash \{1\} : n$ **cannot** be expressed as a product of prime(s)$\} \subseteq \mathbb{Z}^+$.
Suppose $S$ is nonempty.
Then $S$ has a least element $m$ by the Well-ordering Principle.
Since $m$ cannot be a prime, we have

$$m = n_1 n_2 \qquad \text{where } n_1, n_2 \in \mathbb{Z} \text{ with } 1 < n_1, n_2 < m.$$

However, since $m$ is already the least element of $S$, we have $n_1, n_2 \notin S$, so both can be expressed as a product of prime(s).
Hence $m = n_1 n_2$ can also be expressed as a product of prime(s).
This is a contradiction and so $S = \emptyset$.

**Theorem 7.16 (Fundamental Theorem of Arithmetic)**

*Every positive integer $> 1$ can be expressed as a product of prime(s) in a unique way. (Primes are "building blocks" of positive integers $> 1$.)*

**Proof of uniqueness part.**
Let $n \in \mathbb{Z}^+ \backslash \{1\}$ such that

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$$

where $p_i$'s and $q_j$'s are primes, $r, s \in \mathbb{Z}^+$.
Then, by Euclid's lemma,

$$\begin{aligned} p_1 \mid q_1 q_2 \cdots q_s &\implies p_1 \mid q_j \text{ for some } j \\ &\implies p_1 = q_j \quad (\because p_1,\, q_j \text{ are both primes}) \end{aligned}$$

Swapping the indices for $q_j$'s, we have $q_1 = p_1$ and $p_2 p_3 \cdots p_r = q_2 q_3 \cdots q_s$.
Repeat the above, we have $r = s$ and $p_i = q_i$ for $i = 1, 2, \ldots, r$. $\square$

# Fundamental Theorem of Arithmetic

**Direct consequence:**

- Let $a, b \in \mathbb{Z}^+$. Then there are **distinct** primes $p_1, p_2, \ldots, p_r$ such that

$$a = p_1^{m_1} p_2^{m_2} \cdots p_r^{m_r} \quad \text{and} \quad b = p_1^{n_1} p_2^{n_2} \cdots p_r^{n_r}$$

  where $m_1, m_2, \ldots, m_r$, $n_1, n_2, \ldots n_r$ are nonnegative integers. (So the exponent can be zero.)

- Furthermore, $a$ divides $b$ if and only if

$$m_i \leq n_i \quad \text{for all } i = 1, 2, \ldots, r.$$

## Proposition 7.17

*Let $a, b \in \mathbb{Z}^+$ and let $p_1, p_2, \ldots, p_r$ be* ***distinct*** *primes such that*

$$a = p_1^{m_1} p_2^{m_2} \cdots p_r^{m_r} \qquad \textit{and} \qquad b = p_1^{n_1} p_2^{n_2} \cdots p_r^{n_r}$$

*where $m_1, m_2, \ldots, m_r$, $n_1, n_2, \ldots n_r$ are nonnegative integers.*
*Then,*

$$\gcd(a, b) = p_1^{s_1} p_2^{s_2} \cdots p_r^{s_r} \qquad \textit{where } s_i := \min\{m_i, n_i\} \textit{ for } i = 1, 2, \ldots, r.$$

**Proof.**
If $d \mid a$, then $d = p_1^{t_1} p_2^{t_2} \cdots p_r^{t_r}$, where $0 \leq t_i \leq m_i$ for $i = 1, 2, \ldots, r$.
If $d \mid b$, then $d = p_1^{t_1} p_2^{t_2} \cdots p_r^{t_r}$, where $0 \leq t_i \leq n_i$ for $i = 1, 2, \ldots, r$.
So, if $d \mid a$ and $d \mid b$, then

$$d = p_1^{t_1} p_2^{t_2} \cdots p_r^{t_r}, \qquad \text{where } 0 \leq t_i \leq \min\{m_i, n_i\} = s_i \text{ for } i = 1, 2, \ldots, r.$$

For $d$ to be at its greatest, we want the largest possible exponent for each $p_i$'s. The result follows. $\square$

**Direct consequence:** $d$ is a common divisor of $a$ and $b \iff d \mid \gcd(a, b)$.

**Example 7.6**

$$144 = 2^4 \times 3^2 (\times 5^0)$$
$$60 = 2^2 \times 3^1 \times 5^1$$

So

$$\gcd(144, 60) = 2^2 \times 3^1 \times 5^0 = 12.$$

Common divisors of 144 and 60:

$$\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$$

**Proposition 7.18**

*Let $a, b, c \in \mathbb{Z}$ such that $c > 1$. If $c \mid ab$ and $\gcd(a, c) = 1$, then $c \mid b$.*

It is a generalization of Euclid's lemma (put $c = p$ where $p$ is a prime).

**Proof.**
Write

$$a = \prod_{i=1}^{r} p_i^{m_i}, \quad b = \prod_{i=1}^{r} p_i^{n_i}, \quad c = \prod_{i=1}^{r} p_i^{k_i},$$

where $p_1, p_2, \ldots, p_r$ are distinct primes and $m_i, n_i, k_i \in \mathbb{N}$.
Then

$$\begin{aligned} c \mid ab &\implies k_i \leq m_i + n_i \quad \forall i = 1, \ldots, r \\ \gcd(a, c) = 1 &\implies \min\{m_i, k_i\} = 0 \quad \forall i = 1, \ldots, r. \end{aligned}$$

Hence, $k_i \leq n_i \ \forall i = 1, \ldots, r$.
Therefore $c \mid b$. $\square$

**Exercise 7.3**

Let $a, b, c \in \mathbb{Z}$. Suppose $a \mid c, b \mid c$ and $\gcd(a, b) = 1$, show that $ab \mid c$.

# Linear Diophantine Equation

**Question**

Given $a, b, c \in \mathbb{Z}$. Can we find $x, y \in \mathbb{Z}$ such that $ax + by = c$?

It is called a **linear Diophantine equation**.
Equivalently, we want to study the set $\{ax + by : x, y \in \mathbb{Z}\}$.

**Proposition 7.19**

*Given $a, b, c \in \mathbb{Z}$. There exist $x, y \in \mathbb{Z}$ such that $ax + by = c$ if and only if $\gcd(a, b) \mid c$.*

**Proof.**
Let $d = \gcd(a, b)$.
$(\implies)$ Assume there are $x, y \in \mathbb{Z}$ such that $ax + by = c$.
Then $d \mid a, d \mid b \implies d \mid c = ax + by$.
$(\impliedby)$ Assume $d := \gcd(a, b) \mid c$, i.e. $c = dr$ for some $r \in \mathbb{Z}$.
By Bézout's Identity, there exist $m, n \in \mathbb{Z}$ such that $d = am + bn$.
Then $c = dr = amr + bnr = ax + by$ by letting $x = mr, y = nr \in \mathbb{Z}$. $\square$

# Linear Diophantine Equation

Further suppose $0 \neq d := \gcd(a, b) \mid c$ (so the equation $ax + by = c$ has a solution).
Let $u = a/d$, $v = b/d \in \mathbb{Z}$. Then

$$\gcd(u, v) = \gcd(\frac{a}{d}, \frac{b}{d}) = 1$$

**Claim 1:** If $(x, y)$ is a solution, $(x + kv, y - ku)$ is also a solution for any $k \in \mathbb{Z}$.
**Reason:**
$a(x + kv) + b(y - ku) = ax + by + (akv - bku) = c + (dukv - dvku) = c$.
**Claim 2:** If $(x_1, y_1), (x_2, y_2)$ are solutions, then $x_2 = x_1 + kv$, $y_2 = y_1 - ku$ for some $k \in \mathbb{Z}$.
**Reason:**

$$ax_1 + by_1 = c = ax_2 + by_2$$
$$ux_1 + vy_1 = ux_2 + vy_2$$
$$v(y_1 - y_2) = u(x_2 - x_1)$$

Since $\gcd(u, v) = 1$, $v \mid (x_1 - x_2)$, i.e. $x_2 - x_1 = kv$ (or $x_2 = x_1 + kv$).
Then, $v(y_1 - y_2) = ukv$,and so $y_2 = y_1 - ku$.

**Example 7.7**

You are given a glass and a cup which can hold 105 mL and 180 mL, a big bucket and a water source. Can we measure exactly 30 mL of water using them?

Solve $180x + 105y = 30$ where $x, y \in \mathbb{Z}$.
Since $\gcd(180, 105) = 15 \mid 30$, so the above equation is solvable.
By the extended Euclidean algorithm,
$15 = \gcd(180, 105) = 180 \times 3 - 105 \times 5$, and so

$$180 \times 6 - 105 \times 10 = 30.$$

So $(x, y) = (6, -10)$ gives a particular solution.
All solutions are in the form $(6 + 7k, -10 - 12k)$, $k \in \mathbb{Z}$.

Fill the cup with water and pour it into the bucket. Do it 6 times.
Then, pour water form the bucket into the glass and throw away the water in the glass. Do it 10 times.

**Definition 7.5 (Rational and irrationals)**

Let $x \in \mathbb{R}$.

(a) We say that $x$ is a **rational number** if there exist $m, n \in \mathbb{Z}$ such that $n \neq 0$ and $m = nx$ (i.e. $x = \frac{m}{n}$).

(b) We say that $x$ is an **irrational number** if $x$ is not a rational number.

The set of all rational numbers is denoted by $\mathbb{Q}$.

**Theorem 7.20**

*Let $x, y \in \mathbb{Q}$. Then*

(a) $x + y$, $x - y$ *and* $xy \in \mathbb{Q}$;

(b) $\frac{x}{y} \in \mathbb{Q}$ *provided* $y \neq 0$.

**Exercise 7.4**

Give examples that the sum and product of irrational numbers may not be irrational.

**Definition 7.6 (Nonnegative $n$-th root of a nonnegative real number)**

Suppose $n$ is a positive integer, and $a, \rho$ are nonnegative real numbers. Then we say $\rho$ is a **nonnegative $n$-th real root of** $a$ if $\rho^n = a$.

**Theorem 7.21**

*Suppose $n \in \mathbb{Z}^+$ and $a$ is a nonnegative real number. Then there is a unique nonnegative real number $\rho$ such that $\rho^n = a$. This number is denoted by $\sqrt[n]{a}$.*

**Remark:** The existence of $\sqrt[n]{a}$ will be justified in Mathematical Analysis I.

**Example 7.8 (Irrationality of $\sqrt{2}$)**

$\sqrt{2}$ is irrational.

Suppose the contrary.
Then $\sqrt{2} = \dfrac{a}{b}$ for some $a, b \in \mathbb{Z}^+$ that are relatively prime.
Now,

$$\begin{aligned} 2b^2 = a^2 &\implies 2 \mid a^2 \\ &\implies 2 \mid a && \text{(Euclid's lemma)} \end{aligned}$$

Let $a = 2c$ for some $c \in \mathbb{Z}^+$. Then,

$$\begin{aligned} 2b^2 = a^2 = (2c)^2 = 4c^2 &\implies b = 2c^2 \\ &\implies 2 \mid b^2 \\ &\implies 2 \mid b && \text{(Euclid's lemma)} \end{aligned}$$

Hence, $2 \mid a$ and $2 \mid b$, which contradicts the assumption that $a, b$ are relatively prime.
Therefore $\sqrt{2}$ is irrational.

**Exercise 7.5**

Let $a, b, c, d$ be rational numbers. Suppose $a + b\sqrt{3} = c + d\sqrt{3}$. Show that $a = c$ and $b = d$. You may assume that $\sqrt{3}$ is irrational.

**Exercise 7.6 (Irrationality of surds of positive prime numbers)**

Suppose $n$ is an integer greater than 1, and $p$ is a positive prime number. Then $\sqrt[n]{p}$ is an irrational number.

Suppose the contrary.
Then $\sqrt[n]{p} = \dfrac{a}{b}$ for some $a, b \in \mathbb{Z}^+$ that are relatively prime.
Now,

$$\begin{aligned} pb^n = a^n &\implies p \mid a^n \\ &\implies p \mid a && \text{(Euclid's lemma)} \end{aligned}$$

Let $a = pc$ for some $c \in \mathbb{Z}^+$. Then,

$$\begin{aligned} pb^n = a^n = (pc)^n = p^n c^n &\implies b^n = p^{n-1} c^n \\ &\implies p \mid b^n && \text{(note } n > 1\text{)} \\ &\implies p \mid b && \text{(Euclid's lemma)} \end{aligned}$$

Hence, $p \mid a$ and $p \mid b$, which contradicts the assumption that $a, b$ are relatively prime.
Therefore $\sqrt[n]{p}$ is irrational.

**Exercise 7.7**

Let $n \in \mathbb{Z}^+$. Show that $\sqrt{n}$ is rational if and only if $n$ is a perfect square, i.e. $n = m^2$ for some $m \in \mathbb{Z}$.